

Total No. of Questions : 8]

SEAT No. :

PE2476

[Total No. of Pages : 2

[6584]-458

**B.E. (Computer Engineering) (AR & VR) (Honors)
VIRTUAL REALITY IN GAME DEVELOPMENT
(2019 Pattern) (Semester - VII) (410701)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Explain the following terms: **[6]**

- i) Design with virtual environments
- ii) Design for virtual environments
- iii) Design of virtual environments

b) State different tools for building VR applications. **[6]**

c) Explain block diagram for a multimodal VR system developed at HRL Laboratories. **[6]**

OR

Q2) a) Explain Formative usability evaluation cycle. **[6]**

b) Explain in detail Task analysis of Make stage. **[6]**

c) Explain different guidelines of Iterative Design. **[6]**

Q3) a) Explain working with Unity's UI System. **[6]**

- i) UI Canvas
- ii) UI Text and Images
- iii) UI Buttons

b) Explain Interactive NPCs and Non-interactive NPCs. **[6]**

c) Explain working with Sprites in Unity. Also describe how to actually use Sprites in Unity. **[6]**

OR

Q4) a) Explain step to develop town view using unity. **[6]**

b) Differentiate between Animations in Unreal engine Vs Unity Engine. **[6]**

c) Explain the term NPCs and Interactions. **[6]**

P.T.O.

- Q5)** a) How to add background music and sound effect for gaming application using unity? [6]
 b) How to building in-game menu structures? Explain. [6]
 c) How to Implement the turn-based battle system? Explain. [5]

OR

- Q6)** a) Illustrate the overall design process of the Battle System in Unity. [6]
 b) Consider The Following Code Snippet Below: [6]

Class Mover: MonoBehaviour

```
{
Vector3 Target;
Float Speed;
Void Update()
{
}
}
```

Finish This Code So The GameObject Containing This Script Moves With Constant Speed Towards Target, And Stop Moving Once It Reaches 1.0, or Less, Units of Distance?

- c) Illustrate the process to build the shop scene in unity. [5]
- Q7)** a) What are different Questionnaire of Kennedy Simulator Sickness? [5]
 b) State any six system factor that contribute to adverse effects. [6]
 c) What is unified model of motion sickness? Explain with neat diagram. [6]

OR

- Q8)** a) What are different Questionnaire of Kennedy Simulator Sickness? [5]
 b) State different examples of reducing adverse effects. [6]
 c) Explain any three terms w.r.t. physical issues involved with the use of VR equipment. [6]
 i) Physical fatigue
 ii) Headset fit
 iii) Injury and Hygiene

